PRADEEP KALLURI

Cloud Data Infrastructure Engineer | Financial Platforms & Open-Source Systems
https://www.linkedin.com/in/pradeepkalluri/
https://github.com/kalluripradeep

PROFESSIONAL PROFILE

Cloud Data Engineer specialising in scalable financial data platforms, with **four years** of experience across cloud data engineering, fintech infrastructure, open-source software and production analytics systems. I build the data layer behind heavily used banking and digital finance products, enabling secure, reliable processing of customer, payment, transaction and engagement data at scale. Focused on advancing global data engineering best practices by combining Tier-1 digital banking innovation with a growing presence in the wider data engineering ecosystem through merged open-source contributions to widely adopted open-source technologies relied upon across global data ecosystems, invited technical speaking and curated writing, and early-stage startup mentorship.

KEY AREAS OF EXPERTISE

Financial data platform engineering | Cloud-native data infrastructure across AWS and Azure | Distributed pipeline architecture for high-volume banking data | Real-time financial event processing using Kafka and PySpark | Workflow orchestration and production automation using Apache Airflow | Analytics engineering and transformation using dbt and Snowflake | Lakehouse and medallion architecture for scalable reporting | Data quality, lineage and observability for trusted analytics | Performance optimisation of enterprise ETL/ELT systems

INDUSTRY ENGAGEMENT AND OPEN-SOURCE CONTRIBUTIONS

- **Active contributor to widely adopted global open-source data infrastructure projects through GitHub** https://kalluripradeep.github.io/. Recorded 373 contributions in the last year alone to 10 public repositories, gaining 18 stars. My contributions include projects connected to major data engineering frameworks such as Apache Airflow and dbt-core, with multiple contributions merged into core codebases used globally by data engineers to build production-grade data pipelines, automate workflow orchestration, transform raw data into analytics-ready models, and improve the reliability of modern cloud data platforms.
    - **dbt-core open-source project:** Contributed multiple pull requests to the project, collaborating with maintainers through technical discussions, code reviews, and approvals before my contributions were merged into the main branch (https://github.com/dbt-labs/dbt-core/pull/12388, https://github.com/dbt-labs/dbt-core/pull/12232, https://github.com/dbt-labs/dbt-core/pull/12562, https://github.com/dbt-labs/dbt-core/pull/12618)
    - **Apache Airflow open-source framework:** Identified, implemented, and merged a fix for a TaskFlow inference issue in Apache Airflow open-source framework, collaborating with project maintainers, enhancing the framework's reliability, benefitting the global community of developers and data engineers using the framework (https://github.com/apache/airflow/pull/63053, https://github.com/apache/airflow/pull/63110, https://github.com/apache/airflow/pull/61005).
    - **Shared practical guidance** on open-source contributions and engineering best practices based on my contributions, published in the official Apache Airflow publication receiving 74 community claps (https://medium.com/apache-airflow/a-beginners-guide-to-contributing-to-apache-airflow-065fc7081dbc, https://medium.com/apache-airflow/rewriting-my-apache-airflow-pr-when-your-first-solution-isnt-the-right-one-8c4243ca9daf).
    - **Appointed as editor of the official Apache Airflow Medium publication** by an ASF PMC contributor, leveraging the platform to disseminate technical knowledge on workflow orchestration and modern data engineering practices to a global practitioner audience.
- **Mentor | Plug and Play (https://www.plugandplaytechcenter.com/):** Mentor at the Plug and Play innovation ecosystem (one of the world's largest startup accelerator networks spanning 60+ global offices). Provided guidance to 40 early-stage founders on cloud-native architecture, AI/ML infrastructure, data engineering, scalability, automation, and enterprise-grade technology practices to build production-ready digital products. Repeatedly invited as a speaker, technical writer to 71K+ readers, and open source contributor to Apache Airflow and dbt-core
- **Lead Mentor | TiE Hyderabad (https://tiehyderabad.org/):** Served as a Lead Mentor, guiding 5 early-stage startup founders in Hyderabad, providing guidance on cloud-native architecture, data engineering, AI-driven automation, scalability, and technology strategy to help them build production-ready digital products. Helped founders achieve 40% reduction in manual operational effort across 3 startups through workflow automation guidance; 30% improvement in application performance through infrastructure optimisation; 4 startups strengthened investor readiness through improved technical documentation and architectural maturity. Invited to the "Open Mic for Entrepreneurs" event as a speaker on cloud-native engineering, scalable data platforms, and AI automation for startup founders.
- **Used technical writing to translate complex production engineering challenges into practical guidance for global software engineering and data engineering community.**

- Authored an article "*How I Fixed a Silent Production Bug in Apache Airflow That Affected Thousands of Deployments*" explaining how I investigated, fixed and contributed a backward-compatible open-source improvement to Apache Airflow, to help engineers working with it and facing similar challenges. The article was featured on DZone, in April 2026, and has received 2.6K+ views till date (https://dzone.com/articles/how-i-fixed-silent-production-bug-apache-airflow).
- Authored two technical articles on *HackerNoon*, published in March 2026: https://hackernoon.com/does-ensemble-ml-3-ml-models-actually-work-i-tested-it-on-332k-orders on an ensemble machine learning approach, providing practical guidance for deploying production-ready ML systems (186 reads), and https://hackernoon.com/real-time-data-quality-monitor-earns-a-54-proof-of-usefulness-score-by-building-an-open-source-data-observability-dashboard introducing an open-source real-time data observability dashboard built using Apache Kafka and machine learning as a cost-effective alternative to enterprise data observability platforms (249 reads).
- Shared practical guidance on Apache Airflow, data quality, idempotency, and resilient data engineering through a real-world case study of investigating and resolving a production data pipeline incident where the same dataset was processed 47 times, in the article titled "*The Weekend Our Pipeline Processed the Same Data 47 Times*" published on The New Stack (https://thenewstack.io/the-weekend-our-pipeline-processed-the-same-data-47-times/) in Jan 2026.

PROFESSIONAL EXPERIENCE

NatWest (https://www.natwest.com/) is part of NatWest Group, one of the UK's largest retail and commercial banking institutions, serving over 19 million customers across personal, business, and international banking platforms within a heavily regulated financial environment.

Data Engineer | NatWest Bank Cloud Data Engineering, London, UK | September 2024 – Present

- Led the design and implementation of the cloud data infrastructure for **NatWest Accelerator**, the UK's largest entrepreneur support platform, which provides a dedicated mobile application for thousands of business owners to access mentoring, coaching, networking events and business growth resources across 12 accelerator hubs nationwide (https://play.google.com/store/apps/details?id=uk.co.disciplemedia.theaccelerator&hl=en_GB).
  - Built the end-to-end data platform that powered engagement analytics across the Accelerator ecosystem, giving programme leaders real-time visibility into entrepreneur participation and enabling them to optimise programme delivery, make data-driven decisions and strengthen the effectiveness of NatWest's flagship entrepreneurship initiative.
  - Designed the analytical and orchestration layers that transformed engagement data into actionable business insights, and received recognition from UK-based founders for the platform's technical complexity, scalability and impact.
- Designed the data platform powering **NatWest Bankline Mobile application**'s core banking features - **a** production-grade corporate banking platform used by UK businesses to manage accounts and approve high-value payments, with over a 100,000 downloads (https://play.google.com/store/apps/details?id=com.rbs.banklinemobile.natwest).
  - Developed the underlying payment processing, transaction history, analytics, and reporting infrastructure that enabled these customer-facing banking services to operate reliably, securely, and at enterprise scale.
- Designed and maintained the end-to-end data platform powering customer-facing features of the **NatWest International mobile banking application**, with application securing a 100k+ downloads and 4.7 Playstore review.
  - Developed scalable data ingestion pipelines, distributed PySpark processing frameworks, Snowflake data models, and analytics solutions that enabled real-time banking services across multiple countries and regulatory jurisdictions.
  - Enabled critical mobile banking features including Paym integrations, multi-currency banking, instant and secure cross-border transactions, cardless cash withdrawals, customer personalisation settings, and real-time account balance updates.
  - Engineered multi-region Kafka and Amazon S3 ingestion pipelines processing high-volume real-time financial event data, supporting customer account, transaction, and payment data processing across 20+ countries including APAC and EU with high availability and low latency (https://play.google.com/store/apps/details?id=com.rbs.mobile.android.natwestoffshore&hl=en_GB).
  - Designed curated Snowflake data models supporting payment approvals, 15-month transaction history functionality, account personalisation features, and multi-jurisdictional regulatory reporting. Enforced data governance and compliance standards aligned with offshore banking regulations
- Invited twice by the prestigious **Oxford Microsoft Data Platform Group** (a UK professional community for Microsoft data and analytics practitioners and part of the wider Azure Data Community) to share NatWest production data engineering learnings to the wider UK data platform community in my professional capacity as a NatWest Data Engineer. Shared real-world engineering lessons on Kafka, PySpark, Snowflake, cloud-native data pipelines, workflow orchestration, monitoring and production best practices, contributing to knowledge-sharing within the wider UK data engineering community.

Accenture (https://www.accenture.com/gb-en) is a global technology and innovation firm operating in over 120 countries, delivering large-scale cloud, data, and digital transformation programmes for enterprise clients across financial services, healthcare, and technology sectors.

Data Engineer | July 2023 - August 2024

- Contributed to a complex multi-cloud transformation programme managing over 100TB of enterprise data, gaining exposure to enterprise-scale architecture and delivery responsibilities.
- Reduced ETL processing times by 70%, from over 12 hours to under 4 hours, through advanced PySpark optimisation, intelligent dataset repartitioning, and execution configuration tuning in Azure Databricks.
- Delivered approximately $30,000 per month in cloud infrastructure cost savings through systematic workload and compute optimisation across Databricks and Snowflake environments.
- Improved data reliability from 80% to 99.5% through robust quality frameworks, schema evolution strategies, rigorous validation controls, and version-controlled transformation logic.
- Enabled near real-time analytics capabilities contributing to over $1 million in annual revenue uplift through improved data accessibility and operational intelligence.
- Engineered production pipelines supporting 15TB of daily data ingestion with 99.8% uptime.
- Implemented CI/CD pipelines using GitHub Actions and Terraform, automating infrastructure provisioning, deployment configuration, and environment management in line with modern DevOps engineering standards.
- Built reusable dbt transformation layers ensuring consistent, modular, and production-ready business logic across curated enterprise datasets.
- Leveraged Microsoft Fabric for unified analytics workflows, integrating lakehouse storage, data pipelines, and semantic modelling to enhance data accessibility for business stakeholders.

Dpoint Group (https://www.dpointgroup.com/) is a Barcelona-based data and analytics practice specialising in business intelligence, SAP data migrations, and cloud-based reporting solutions for European markets.

Data Engineer | Barcelona, Spain | May 2022 - June 2023

- Transformed legacy SAP BW environments to modern Azure-based data architectures. Developed and maintained SSIS-based ETL pipelines extracting and loading data, securely, into Azure-based storage and reporting systems.
- Re-architected on-premise ETL processes to Azure Data Factory, enhancing scalability and reducing maintenance overhead. Built interactive Power BI dashboards supporting executive insights, operational analytics, and KPI monitoring across business functions.
- Automated recurring reporting workflows using Python and Excel VBA, improving delivery speed and reducing manual workload.

CERTIFICATIONS

- PlatformCon 2026 Virtual - 2026
- SnowPro Core Certification: Snowflake - 2026
- Microsoft Certified: Fabric Data Engineer Associate - 2026
- Quantium - Data Analytics Job Simulation: Forage - 2025

EDUCATION

- Master of Business Administration (MBA) | York St John University, London, United Kingdom (Part-Time) | 2023 - 2024
- Bachelor of Internet & Communication Technology | Tor Vergata University, Rome, Italy | 2020 - 2023

HOBBIES & INTERESTS

DataOps & MLOps Best Practices | Modern Data Stack Implementation | Open Source Contributions | Cross-Cultural Engagement | Sport and Team Leadership